

Midterm 1

● Graded

Student

Lama Alsurakhi

Total Points

82 / 100 pts

Question 1

Question 1

18 / 30 pts

1.1 a

12 / 15 pts

– 0 pts Correct

– 3 pts first one is wrong

✓ – 3 pts second is wrong

– 3 pts third is wrong

– 3 pts fourth is wrong

– 3 pts fifth is wrong

1.2 b

6 / 7 pts

– 0 pts Correct

✓ – 1 pt unit is wrong (should be 6000 kJ)

– 1 pt minutes to seconds conversion

– 1 pt watts= joules/seconds

– 7 pts not done or wrong

1.3 c

0 / 8 pts

– 0 pts Correct

– 0 pts Pressure = ρgh

✓ – 8 pts not done or wrong

– 1 pt unit is wrong

– 1 pt ρ is wrong

– 1 pt needs to determine the difference

Question 2

Question 2

32 / 35 pts

2.1 — a

2 / 2 pts

✓ - 0 pts Correct

2.2 — b

3 / 4 pts

- 0 pts Correct

✓ - 1 pt 3-1 is wrong

- 4 pts wrong or not done

- 1 pt not water, this is air. Not dome drawing

- 1 pt x axis should be specific volume, not volume.

2.3 — c,d

5 / 5 pts

✓ - 0 pts Correct

- 1 pt closed system

- 1 pt $\Delta U = Q - W$

2.4 — e

14 / 15 pts

- 0 pts Correct

- 5 pts W1-2 is wrong

- 4 pts W2-3 is wrong

- 6 pts W3-1 is wrong

- 1 pt W1-2 should be negative

- 2 pts n value for polytropic process is wrong

- 2 pts unit conversion

- 1 pt W3-1 should be positive

✓ - 1 pt W3-1 unit conversion

- 3 pts W3-1 should be calculated by natural log

- 1 pt W 3-1 calculation error

- 1 pt calculation error W1-2

2.5 

8 / 9 pts

– 0 pts Correct

✓ – 1 pt calculation error

– 9 pts not done or wrong

Question 3

Question 3

32 / 35 pts

3.1 — a

2 / 2 pts

✓ - 0 pts Correct

3.2 — b

4 / 6 pts

- 0 pts Correct

- 3 pts domes are missing on the graphs

- 5 pts states are not labeled on the graph

- 1 pt state 1 is wrong on Pv graph

- 1 pt state 1 is wrong on Tv graph

- 1 pt state 2 is wrong on Pv graph

- 1 pt state 2 is wrong on Tv graph

✓ - 1 pt state 3 is wrong on Pv graph

✓ - 1 pt state 3 is wrong on Tv graph

3.3 — c,d

5 / 5 pts

✓ - 0 pts Correct

- 5 pts not done or wrong

- 1 pt closed system

3.4 — e

9 / 10 pts

- 0 pts Correct

- 10 pts not done or wrong

- 6 pts W2-3 is not done or wrong (should be constant P)

- 1 pt calculation error

✓ - 1 pt W2-3 should be negative

- 1 pt P is wrong

- 2 pts v3 is wrong

- 2 pts v2 is wrong

3.5 f

10 / 10 pts

✓ - 0 pts Correct

- 1 pt calc erro

- 3 pts x not calculated or wrong

- 3 pts u1 is wrong

- 10 pts not done

- 1 pt u3 is wrong

- 2 pts deltaU calculations are wrong

3.6 g

2 / 2 pts

✓ - 0 pts Correct

- 1 pt state 1 is wrong

- 1 pt state 2 is wrong

- 2 pts not done or wrong

- 0.5 pts state 3 is wrong

NAME: Lama AlsurakhiPlease fill in answers in the space supplied and show all your work to receive full credit.1. a) (15 pts) State whether the following statements are **True or False**

- F An ideal gas at a given state expanded to a fixed final volume at constant pressure or at constant temperature. Value of work is smaller in constant pressure case.
- F Seven people stand on a scale and have a mass of 1280 lb. A specific property called the pep is defined as the number of people divided by mass. Pep is an intensive property. $\rightarrow$ dependant on mass
- F Isochoric process means no heat transfer. constant volume
- F A two-phase liquid-vapor mixture with equal volumes of saturated vapor and saturated liquid has a quality more than 0.5.
- F Energy transferred by heat to a closed system can be calculated by the area under the curve in P-V diagram. $\rightarrow$ work

b) (7pts) (FE Exam)

A 2-kW electric resistance heater in a room is turned on and kept on for 50 min. Determine the amount of energy transferred.

$$\frac{2 \text{ kg} \cdot \text{m}}{\text{s}} \times \frac{60 \text{ sec}}{1 \text{ min}} \times 50 \text{ min} = \boxed{6000 \text{ kg} \cdot \text{m}}$$

$$\text{kW} = \frac{\text{kg} \cdot \text{m}}{\text{s}}$$

$$N = \frac{\text{kg} \cdot \text{m}}{\text{s}^2}$$

$$E = J = N \cdot \text{m}$$

c) (8pts) (FE Exam)

Consider a fish swimming 5 m below the free surface of water. Determine the increase in the pressure exerted on the fish when it dives to a depth of 25 m below the free surface.

$$\Delta V = 25 - 5 = 20 \text{ m}$$

$$Pv = nRT$$

$$P = \frac{nRT}{V} = \frac{1}{V} = \frac{1}{20}$$

The Pressure increase by $\frac{1}{20}$

2. (35 points)

A gas in a piston-cylinder assembly (7kg) at 5 bar and 15 m^3 undergoes a cycle with three processes in series.

Process 1-2: Isobaric compression to 10 m^3 .

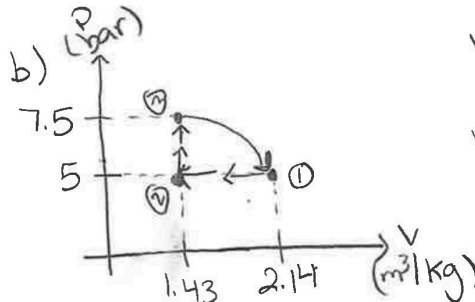
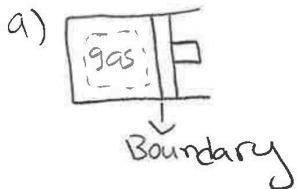
Process 2-3: Isochoric increase of pressure to 7.5 bar.

Process 3-1: Polytropic expansion to initial state.

Kinetic and potential effects can be ignored.

- (2 pts) Draw a schematic of the system and label the system boundary.
- (4 pts) Carefully sketch the process on a P-v diagram (Pressure-specific volume)
- (3 pts) State the engineering model
- (2 pts) Write the simplified energy balance
- (15 pts) Calculate the work transferred in kJ for each step. Is work done on the system or by the system?
- (9 pts) Calculate the overall heat transferred in kJ. Is heat transferred to the system or from the system?

+B
-Q
+T
-F



$$v_1 = \frac{15 \text{ m}^3}{7 \text{ kg}} = 2.14 \text{ m}^3/\text{kg}$$

$$v_2 = \frac{10 \text{ m}^3}{7 \text{ kg}} = 1.43 \text{ m}^3/\text{kg}$$

c) Closed system
 $\Delta PE = \Delta KE = 0 \text{ kJ}$

d)

$$\Delta PE + \Delta KE + \Delta u = Q - W$$

$$\Delta u = Q - W \quad \Delta E = \text{Isobaric state} + \text{Polytropic state}$$

e) Work:

1-2:

$$W = \int P dv = 5 \text{ bar} \int_{15}^{10} dv = 5 \text{ bar} (10 - 15) \text{ m}^3 = -25 \text{ bar} \cdot \text{m}^3 \times \frac{100 \text{ kJ}}{1 \text{ bar} \cdot \text{m}^3} = -2500 \text{ kJ}$$

2-3:

$$W = \int P dv = 0 \text{ kJ} \quad \text{Constant volume: } (10 - 10) = 0$$

3-1:

$$Pv^n = \text{Constant} \quad P_3 v_3^n = P_1 v_1^n \quad (7.5 \text{ bar})(10 \text{ m}^3)^n = (5 \text{ bar})(15 \text{ m}^3)^n \quad n = \log_{3/2} 1.5$$

$$P_3 v_3' = \text{Constant} \quad (7.5)(10) = 75 \quad PV = 75 \quad P = 75/V$$

$$W = \int_{10}^{15} \frac{75}{v} dv = 75 \ln v \Big|_{10}^{15} = 30.41 \text{ kJ}$$

$$W_{\text{total}}: -2500 \text{ kJ} + 0 \text{ kJ} + 30.41 \text{ kJ} = -2469.6 \text{ kJ}$$

work being done on the system

$$f) \Delta u = Q - w$$

↓

0 went back to state 1

$$Q = w$$

$$Q = -2469.6 \text{ kJ}$$

Heat being transferred from the system

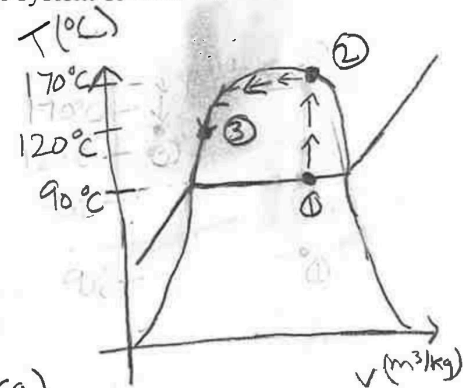
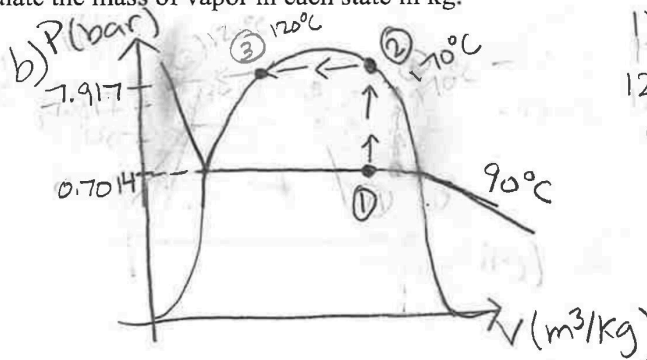
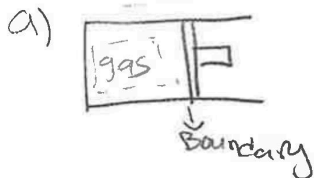
3. (35 points)

5 kg of liquid-vapor mixture of water contained in a piston-cylinder assembly undergoes a process from an initial state where the temperature is 90°C.

- **Process 1-2** System is heated at constant volume to a saturated vapor state at 170°C.
- **Process 2-3** System is cooled at constant pressure to 120°C.

Kinetic and potential effects can be ignored.

- (2 pts) Draw a schematic of the system and label the system boundary.
- (6 pts) Carefully sketch the process on a:
 - P-v diagram (Pressure-specific volume)
 - T-v diagram (Temperature-specific volume)
- (3 pts) State the engineering model
- (2 pts) Write the simplified energy balance
- (10 pts) Calculate the work transferred in kJ. Is work done on the system or by the system?
- (10 pts) Calculate the heat transferred in kJ. Is heat transferred to the system or from the system?
- (2 pts) Calculate the mass of vapor in each state in kg.



c) Closed system
 $\Delta E = \Delta KE = 0$

d) $\Delta KE + \Delta PE + \Delta U = Q - W$
 $\Delta U = Q - W$ $\Delta E = \text{Isobaric State}$

e) work:

1-2

$W = \int P dv = \boxed{0 \text{ kJ}}$ constant volume

2-3

$W = \int P dv = P(u_g - u_f) \cdot m = 7.917 \text{ bar} (0.2428 \text{ m}^3/\text{kg} - 1.0603 \times 10^{-3} \text{ m}^3/\text{kg}) \cdot 5 \text{ kg}$

$W = 9.569 \text{ bar} \cdot \text{m}^3 \times \frac{100 \text{ kJ}}{1 \text{ bar} \cdot \text{m}^3} = \boxed{956.9 \text{ kJ}}$

$$W_{\text{total}} = 0 + 956.9 \text{ kJ} = \boxed{956.9 \text{ kJ}}$$

work being done by the system

f) 2-3

$$\Delta u = Q + W$$

$$m(u_3 - u_2) = Q - W$$

$$5 \text{ kg}(503.50 - 2576.5) \text{ kJ/kg} = Q - 956.9 \text{ kJ}$$

$$\boxed{Q_{2-3} = -9408.07 \text{ kJ}}$$

1-2

$$\Delta u = Q - W$$

$$m(u_2 - u_1) = Q$$

$$5 \text{ kg}(2576.5 - 593.79) = Q$$

$$\boxed{Q_{1-2} = 9913.55 \text{ kJ}}$$

$$u_1 = (1-x)u_f + u_g x \quad \text{at } 90^\circ\text{C}$$

$$u_1 = (1-0.102)(376.85) + (2441.5)(0.102)$$

$$u_1 = 593.79$$

$$V_1 = V_2 = V = 0.2428 \quad \text{at } 170^\circ\text{C}$$

$$x = \frac{V - v_f}{v_g - v_f} \quad \text{at } 90^\circ\text{C}$$

$$x = \frac{(0.2428) - (1.0360 \times 10^{-3})}{(2.361) - (1.0360 \times 10^{-3})}$$

$$x = 0.102$$

$$Q_{\text{total}} = -9408.07 + 9913.55 \text{ kJ} = \boxed{505.48 \text{ kJ}}$$

Heat being transferred to the system

e) State 1: $m_{\text{vapor}} = x(m_{\text{total}}) = 0.102(5 \text{ kg}) = \boxed{0.51 \text{ kg}}$

State 2: $\boxed{100\% \text{ vapor}}$

State 3: $\boxed{0\% \text{ vapor}}$